

eBPF Programming with Rust

Pipeline · libbpf · Aya Framework · DMA Latency Demo

Your Name

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IC Design Division

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1. eBPF — A Quick Refresher

1.1 What is eBPF?

The model in one sentence

eBPF lets you load user-supplied programs into the kernel **without a kernel patch, without a module, and without rebooting** — the kernel verifier guarantees safety.

The four-step contract

1. **Write** — BPF bytecode (from C or Rust source)
2. **Verify** — kernel verifier: bounded loops, no OOB, type-checked
3. **JIT** — native machine code; zero interpreter overhead after load
4. **Attach** — hook point (kprobe, tracepoint, XDP, LSM, ...) fires on event

Hook types your team uses

kprobe / kretprobe	any kernel function entry/return
tracepoint	stable kernel trace points
tp_btf	typed tracepoints — CO-RE friendly
perf_event	hardware PMU counters
xdp	NIC fast path — pre network stack
lsm	Linux Security Module hooks
cgroup_skb	per-cgroup packet filtering

1.1 What is eBPF?

If the verifier accepts the program, it **cannot** crash the kernel, infinite-loop, or access out-of-bounds memory. This is a formal proof, not a heuristic.

raw_tp

raw tracepoints — lowest overhead

1.2 eBPF maps — the data bridge

Maps = the only I/O channel for BPF programs

- Shared memory between kernel BPF code and userspace reader
- Created by the loader before the program is attached
- Accessed from both sides via file descriptors

Type	Typical use
HASH	key → value lookup
RINGBUF	high-throughput event stream ✓
PERCPU_ARRAY	per-CPU stats, lock-free
LRU_HASH	connection tracking

Ring buffer — the preferred choice

Introduced: Linux 5.8 — `BPF_MAP_TYPE_RINGBUF`

- Variable-length records — no fixed-size overhead
- Single contiguous allocation — cache-friendly
- `epoll` / `AsyncFd` compatible — Tokio-native in Aya
- Dropped-event counter exposed to userspace for monitoring
- **In Aya:**

```
#[map] static EVENTS: RingBuf = RingBuf::with_byte_size(4 * 1024 * 1024, 0);
```

1.2 eBPF maps — the data bridge

PERF_EVENT_ARRAY	legacy event pipe
ARRAY	fixed-size indexed data

Prefer `RINGBUF` over `PERF_EVENT_ARRAY` for all new work — lower overhead, simpler consumer, no per-CPU complexity.

2. Does Rust Fit eBPF?

2.1 Why Rust is a natural fit for eBPF programs

Shared constraints

eBPF programs and Rust both operate under the same fundamental constraint:

no undefined behaviour is acceptable.

- The BPF verifier rejects programs it cannot prove safe
- Rust's type system rejects programs it cannot prove safe
- Both operate at compile time or load time — not at runtime

no_std alignment

eBPF programs run with no kernel library, no allocator, no OS primitives. Rust's #!

[no_std] mode is the natural target — aya-

Type safety across the kernel boundary

The most expensive eBPF bug class in C: a map struct definition in the BPF program and the userspace loader that silently diverge.

```
// BPF program (C):  
struct event { u64 ts; u32 pid; };  
bpf_ringbuf_output(&rb, &e, sizeof(e), 0);  
  
// Userspace loader (C) — different file:  
struct event { u64 ts; u64 pid; }; // u64 ≠  
u32 !  
// Reads wrong data — no compile error, no  
warning
```

Aya's solution: one #[no_std] **common crate**, compiled for both targets. The struct is

2.1 Why Rust is a natural fit for eBPF programs

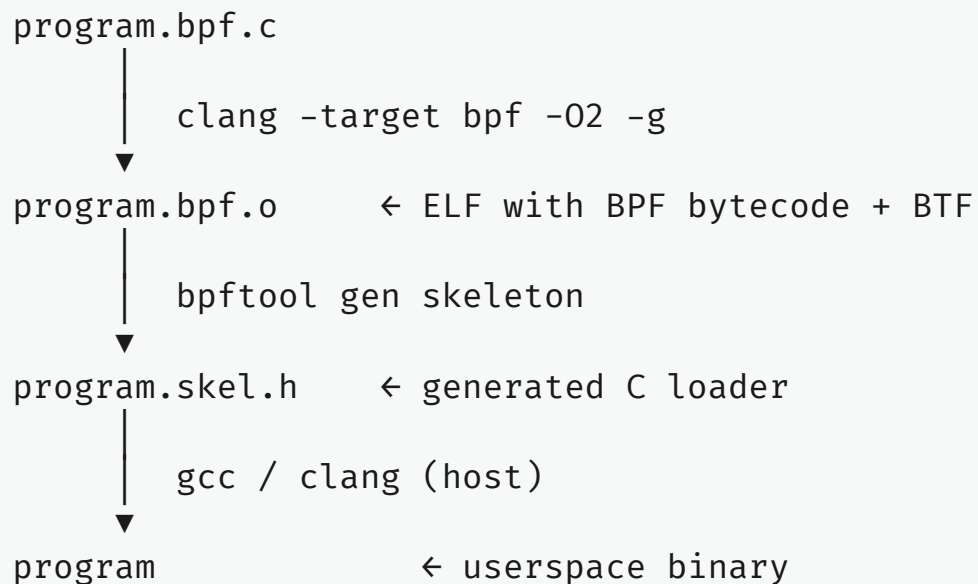
`ebpf` provides the BPF-side runtime
(`bpf_helpers`, map types, program macros)
without any standard library dependency.

defined **once**. Layout disagreement is a
compile error, not a runtime bug.

This is the highest-value safety property
of Rust for eBPF — not just memory
safety in the BPF program, but **type-safe
communication across the kernel/
userspace boundary**.

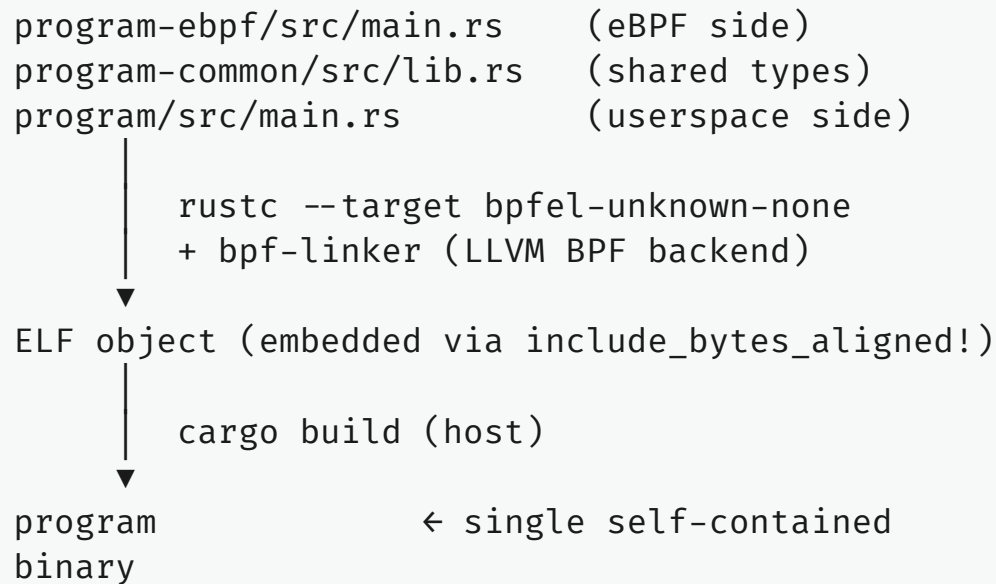
2.2 The bytecode generation question

C eBPF toolchain



What you need: clang + LLVM (BPF backend) + bpftool + libelf + libbpf. C toolchain mandatory for the BPF program even if userspace is Rust (libbpf-rs).

Rust / Aya toolchain



What you need: rustc (nightly) + bpf-linker. **No clang, no bpftool, no libelf, no C toolchain.** The BPF ELF is embedded in the userspace binary — one artifact to deploy.

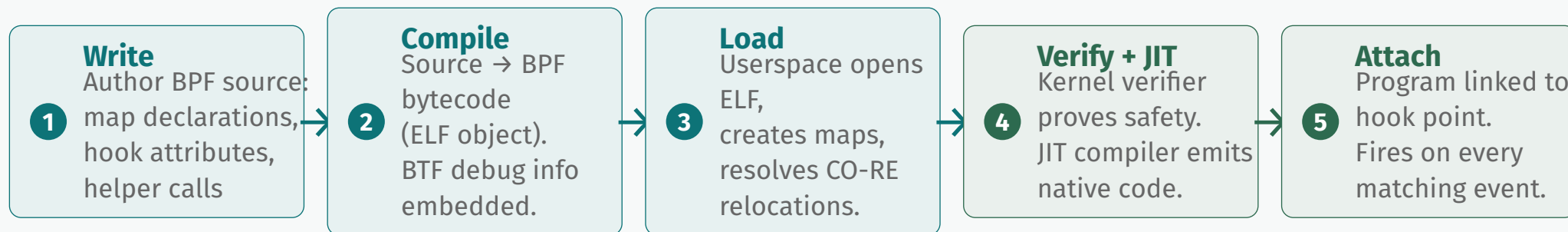
2.2 The bytecode generation question

Aya does not wrap libbpf — it is a **pure-Rust reimplementation** of the BPF syscall layer, built on `libc` only.

github.com/aya-rs/aya — “built from the ground up purely in Rust”

3. The eBPF Pipeline — Source to Running Hook

3.1 Five stages from source to attached program



Stage 1 — Write

- Declare maps at crate/file level with attributes
- Annotate functions with hook type (`#[kprobe]`, `#[tracepoint]`, ...)
- Call BPF helpers (`bpf_ktime_get_ns`, `bpf_get_current_comm`, ...)
- Access kernel struct fields via CO-RE macros

Stage 5 — Attach (detail)

The attachment is a **file descriptor** that the kernel holds. When it closes, the program is automatically detached — no cleanup code needed.

- `kprobe`: attach to any kernel function symbol by name

3.1 Five stages from source to attached program

- Share event struct definition with userspace via common crate / header
- `tracepoint`: attach to `subsystem/event_name` in `tracefs`
- `xdp`: attach to a network interface by name
- `lsm`: attach to a specific LSM hook name
- **Aya**: the attachment is a Rust struct implementing `Drop` — RAI cleanup guaranteed, even on panic paths.

4. The libbpf Approach — Stage by Stage

4.1 libbpf: write & compile

Stage 1 — Write (C)

```
// program.bpf.c
#include "vmlinux.h"           // all kernel
types in one header
#include <bpf/bpf_helpers.h>
#include <bpf/bpf_tracing.h>

// Map declaration – section name encodes type
struct {
    __uint(type, BPF_MAP_TYPE_RINGBUF);
    __uint(max_entries, 4 * 1024 * 1024);
} events SEC(".maps");

// Shared event struct – must match userspace
manually
struct dma_event { u64 start_ns; u32 nents; u32
pid; };

// Hook attribute – clang section magic
SEC("kprobe/dma_map_sg")
int BPF_KPROBE(dma_map_sg_enter, ...) {
    struct dma_event e = {};
```

Stage 2 — Compile (C toolchain)

```
# Generate vmlinux.h – all kernel types from BTF
bpftool btf dump file /sys/kernel/btf/vmlinux \
    format c > vmlinux.h

# Compile BPF source to BPF ELF object
# -target bpf: BPF bytecode output
# -g:          embed BTF debug info (CO-RE)
# -O2:         required: some BPF features need
optimisation
clang -target bpf -g -O2 \
    -D__TARGET_ARCH_x86 \
    -I. \
    -c program.bpf.c \
    -o program.bpf.o

# Inspect the result
llvm-objdump -d program.bpf.o # disassemble
BPF bytecode
bpftool btf dump file program.bpf.o # verify
BTF is present
```

4.1 libbpf: write & compile

```
e.start_ns = bpf_ktime_get_ns();  
// ...  
bpf_ringbuf_reserve(&events, sizeof(e), 0);  
return 0;  
}  
char LICENSE[] SEC("license") = "GPL";
```

The output `program.bpf.o` is a standard ELF file containing:

- `.text` section: BPF bytecode instructions
- `.BTF` section: type information for CO-RE
- `.maps` section: map definitions
- Relocation sections for helper calls

4.2 libbpf: skeleton generation & loading

Stage 2b — Generate skeleton (optional but recommended)

```
# Auto-generate a type-safe C loader header
bpftool gen skeleton program.bpf.o \
    > program.skel.h
```

The generated `program.skel.h` provides:

```
struct program_bpf {
    struct bpf_object_skeleton *skeleton;
    struct bpf_object *obj;
    struct {
        struct bpf_map *events; // typed map
access
    } maps;
    struct {
        struct bpf_program *dma_map_sg_enter;
    } progs;
    struct {
        struct bpf_link *dma_map_sg_enter;
    } links;
```

Stage 3 — Load & Stage 4 verify (C userspace)

```
#include "program.skel.h"

struct program_bpf *skel;

// Open: parse ELF, discover maps and programs
skel = program__open();

// (optional) pre-load configuration:
skel->rodata->min_latency_ns = 1000;

// Load: create maps, CO-RE relocations,
//         submit to kernel verifier
program__load(skel);
// At this point: kernel has verified and JIT'd
// the program.
// Maps are created and their fds are in skel-
// >maps.*

// Attach: link programs to hook points
```

4.2 libbpf: skeleton generation & loading

```
};  
// Auto-generated lifecycle functions:  
// program__open()  program__load()  
// program__attach() program__destroy()
```

```
program__attach(skel);  
// dma_map_sg_enter now fires on every  
dma_map_sg() call
```

CO-RE relocation happens inside `__load()`:
libbpf reads `/sys/kernel/btf/vmlinux`, patches
field offsets in the BPF bytecode to match
the running kernel's struct layout.

The skeleton collapses open + load +
attach into three typed function calls.
Without skeleton: you call
`bpf_object__open()`, iterate programs, call
`bpf_program__load()` per program, then
`bpf_program__attach()` — verbose and
untyped.

4.3 libbpf: maps & teardown

Maps from the C userspace side

```
// Access map fd from skeleton
int map_fd = bpf_map__fd(skel->maps.events);

// Ring buffer consumer (callback model)
struct ring_buffer *rb = ring_buffer__new(
    map_fd, handle_event, NULL, NULL);

// Poll - blocks until events or timeout
while (running) {
    ring_buffer__poll(rb, 100 /* timeout ms */);
}

// Callback invoked per event
static int handle_event(void *ctx,
    void *data, size_t size) {
    struct dma_event *e = data;
    printf("pid=%u latency=%.3f µs\n",
        e->pid,
        (e->end_ns - e->start_ns) / 1000.0);
    return 0;
}
```

Stage 5 teardown (C)

```
// Teardown - must be called explicitly
ring_buffer__free(rb);
program__detach(skel); // closes bpf_link fds
program__destroy(skel); // frees maps,
                        // programs, object

// If __destroy() is not called:
// - Maps leak until process exits
// - Programs may stay attached if link pinned
//   to bpfes
// - No compiler warning, no safety net
```

Deployment dependencies

To ship a libbpf-based tool you need:

- libbpf.so (or static libbpf.a) on the target
- libelf.so (required by libbpf)
- libz.so (required by libelf)

4.3 libbpf: maps & teardown

- The `program.bpf.o` object file **or** embed it via skeleton

On Android or embedded targets: managing `.so` dependencies across OEM kernel variants is a real distribution problem. This is exactly what Aya solves.

5. How Rust (Aya) Handles Each Stage

5.1 Stage-by-stage comparison

Stage	libbpf (C)	Aya (Rust)
1 — Write BPF source	C source (.bpf.c), SEC() macros, bpf_helpers.h, vmlinux.h	Rust source (no_std), #[kprobe] / #[map] attributes, aya-ebpf crate
2 — Compile → bytecode	clang -target bpf -g -O2 → program.bpf.o	rustc --target bpfel-unknown-none + bpf-linker → ELF object
2b — Skeleton / embedding	bpftool gen skeleton → program.skel.h Included in userspace C source	include_bytes_aligned!() embeds ELF into userspace binary at compile time
3 — Open / parse ELF	bpf_object__open() or skel__open() Reads ELF, discovers maps & progs	Ebpf::load(BYTES) — pure Rust ELF parser (aya-obj), no libelf dep
4 — Load + verify	bpf_object__load() or skel__load() Maps created, CO-RE patches applied, bpf() syscall → verifier → JIT	prog.load() per program type Aya performs CO-RE, calls bpf() syscall Same kernel verifier & JIT path

5.1 Stage-by-stage comparison

4b — Attach to hook

`skel__attach()` or `bpf_program__attach()`
`bpf_link` fd returned

`prog.attach("dma_map_sg", 0)?`
Returns typed handle implementing
`Drop`

5 — Maps: read events

`ring_buffer__new()` + `ring_buffer__poll()`
Callback-based consumer

`RingBuf::try_from(map)?` + `AsyncFd`
`Async` / `epoll` — Tokio-native

6 — Teardown

`skel__detach()` + `skel__destroy()`
Manual, must not forget

RAII — all handles `Drop` automatically
Compiler guarantees cleanup

5.2 Aya BPF-side: write & declare

dma-tracer-ebpf/src/main.rs — BPF program (Rust)

```
#![no_std]
#![no_main]
use aya_ebpf::{
    helpers::{bpf_ktime_get_ns,
               bpf_get_current_pid_tgid},
    macros::{kprobe, kretprobe, map},
    maps::{HashMap, RingBuf},
    programs::{ProbeContext, RetProbeContext},
};
use dma_tracer_common::DmaEvent; // ← shared type

// — Maps — declared at crate level —
#![map]
static START: HashMap<u64, u64> =
    HashMap::with_max_entries(4096, 0);

#![map]
static EVENTS: RingBuf =
    RingBuf::with_byte_size(4 * 1024 * 1024, 0);

// — Hook: kprobe on dma_map_sg entry —
#![kprobe]
pub fn dma_map_sg_enter(ctx: ProbeContext) → u32 {
    let pid_tgid = bpf_get_current_pid_tgid();
    let pid = (pid_tgid & 0xFFFF_FFFF) as u32;
    let key = pid as u64;
    let ts = unsafe { bpf_ktime_get_ns() };
    let _ = START.insert(&key, &ts, 0);
```

kretprobe — measure & emit

```
#![kretprobe]
pub fn dma_map_sg_exit(ctx: RetProbeContext) → u32 {
    let end = unsafe { bpf_ktime_get_ns() };
    let pid = (bpf_get_current_pid_tgid()
               & 0xFFFF_FFFF) as u32;
    let key = pid as u64;

    if let Some(start) = START.get(&key) {
        let _ = START.remove(&key);
        // Reserve slot in ring buffer
        if let Some(mut buf) =
            EVENTS.reserve::<DmaEvent>(0)
        {
            unsafe {
                (*buf.as_mut_ptr()).latency_ns =
                    end.saturating_sub(*start);
                (*buf.as_mut_ptr()).pid = pid;
            }
            buf.submit(0);
        }
    }
    0
}

#![cfg(not(test))]
#![panic_handler]
fn panic(_: &core::panic::PanicInfo) → ! {
```

5.2 Aya BPF-side: write & declare

```
}  
  0
```

```
loop {} // BPF panic = halt  
}
```

5.3 Aya userspace: load, attach, consume

dma-tracer/src/main.rs — userspace loader

```
use aya::{include_bytes_aligned, Ebpf,
          maps::RingBuf,
          programs::{KProbe, KRetProbe}};
use aya_log::EbpfLogger;
use tokio::io::unix::AsyncFd;

// BPF ELF embedded at compile time — no runtime file I/O
static BPF_CODE: &[u8] = include_bytes_aligned!(
    concat!(env!("OUT_DIR"),
            "/dma-latency-tracer-ebpf")
);

#[tokio::main]
async fn main() → anyhow::Result<()> {
    // — Load —————
    let mut bpf = Ebpf::load(BPF_CODE)?;
    EbpfLogger::init(&mut bpf).ok();

    // — Attach kprobe —————
    let entry: &mut KProbe = bpf
        .program_mut("dma_map_sg_enter")?
        .try_into()?;
    entry.load()?;
    entry.attach("dma_map_sg", 0)?;

    // — Attach kretprobe —————
    let exit: &mut KRetProbe = bpf
        .program_mut("dma_map_sg_exit")?
```

Async ring-buffer consumer

```
// — Consume ring buffer —————
let rb_map = bpf.take_map("EVENTS")?;
let ring = RingBuf::try_from(rb_map)?;
// Wrap in AsyncFd → Tokio epoll integration
let async_fd = AsyncFd::new(ring)?;

loop {
    tokio::select! {
        _ = tokio::signal::ctrl_c() => break,
        guard = async_fd.readable() => {
            let mut g = guard?;
            let rb = g.get_inner_mut();
            while let Some(item) = rb.next() {
                let ev: DmaEvent = unsafe {
                    *(item.as_ptr()
                      as *const DmaEvent)
                };
                println!(
                    "pid={} lat={:.2}µs",
                    ev.pid,
                    ev.latency_ns as f64/1000.0
                );
            }
            g.clear_ready();
        }
    }
}

// — Teardown: automatic via Drop —————
```

5.3 Aya userspace: load, attach, consume

```
    .try_into()?;  
    exit.load()?;  
    exit.attach("dma_map_sg", 0)?;
```

```
    // bpf, entry, exit, rb all drop here –  
    // links closed, maps freed, no leak possible  
    Ok::<(), ()  
}
```

6. Aya Architecture Overview

6.1 Component map

Aya crate family

Your userspace loader (std, async) uses: aya, aya-log, aya-obj	
aya Ebpf::load()	– core loader library programs maps links
aya-obj Pure Rust ELF parser,	– ELF + BTF + CO-RE relocation engine
aya-log Reads aya-log-ebpf ring messages	– userspace log receiver

Your BPF program (no_std, no_main) uses: aya-ebpf, aya-log-ebpf	
aya-ebpf #[map] #[kprobe] #[xdp] helpers	– BPF-side runtime

What each component handles

aya	Load BPF ELF, create maps, attach programs, manage links. Pure-Rust bpf() syscall wrapper. CO-RE via aya-obj.
aya-obj	Parse BPF ELF, process BTF sections, apply CO-RE relocations. No libelf dependency.
aya-ebpf	BPF-side runtime: #[map], #[kprobe], #[xdp], #[lsm], ... macros; helper function wrappers; map type structs.

6.1 Component map

<code>aya-log-ebpf</code> – log from BPF programs <code>info!()</code> <code>warn!()</code> → <code>aya-log</code> ring buffer
<code>aya-ebpf-bindings</code> – kernel type defs Generated from kernel uapi headers

`aya-log`

Userspace receiver: polls a dedicated ring buffer for log records from `aya-log-ebpf`.

`aya-log-ebpf`

BPF-side: `info!()`, `warn!()`, `debug!()` macros that format and submit log events to the log ring buffer.

`aya-tool`

CLI: generates Rust bindings (`vmlinux.rs`) from kernel BTF — equivalent of `bpftool btf dump ... format c > vmlinux.h`.

No libbpf, no libelf, no C toolchain required at runtime. The entire stack is

6.1 Component map

pure Rust + one `bpf()` syscall.

github.com/aya-rs/aya, 2024

7. Creating an Aya Project

7.1 Prerequisites and scaffold

Prerequisites

1. Rust nightly (build-std required for BPF target)

```
rustup toolchain install nightly \
    --component rust-src
```

2. bpf-linker – LLVM BPF backend for rustc

```
cargo install bpf-linker
```

3. aya-tool – BTF → Rust bindings generator

```
cargo install aya-tool
```

4. Verify LLVM BPF backend is present

```
rustc +nightly --print target-list | grep bpfel
```

bpfel-unknown-none ← must appear

Scaffold with the Aya template

cargo-generate scaffolds the three-crate workspace

```
cargo install cargo-generate
```

Generated workspace structure

```
dma-latency-tracer/
├── Cargo.toml                ← workspace root
├── rust-toolchain.toml      ← pins nightly
├── dma-latency-tracer-common/
│   └── src/lib.rs           ← #[no_std]
├── shared types
│   └── DmaEvent,
├── histogram bounds
├── dma-latency-tracer-ebpf/
│   ├── .cargo/config.toml  ← target=bpfel-
│   └── unknown-none
│       ├── Cargo.toml      ← aya-ebpf dep
│       └── src/main.rs     ← #[kprobe],
│                               #[kretprobe]
└── dma-latency-tracer/
    ├── build.rs             ← KEY
    └── eBPF crate,         ← cross-compile
                              copies to
```

7.1 Prerequisites and scaffold

```
cargo generate --git \  
  https://github.com/aya-rs/aya-template \  
  --name dma-latency-tracer
```

```
$OUT_DIR  
├── Cargo.toml           ← aya dep, tokio  
└── src/main.rs         ← Ebpf::load,  
attach, ringbuf
```

No xtask needed. A single `cargo build` cross-compiles the eBPF crate (via `build.rs`) and embeds the result. One command, one binary output.

7.2 The three Cargo.toml files — key dependencies

dma-latency-tracer-ebpf/Cargo.toml

```
[package]
name     = "dma-latency-tracer-ebpf"
version  = "0.1.0"
edition  = "2021"

[[bin]]
name     = "dma-latency-tracer-ebpf"
path     = "src/main.rs"

[dependencies]
# BPF-side runtime: macros, map types, helpers
aya-ebpf    = "0.1"
# Logging from BPF programs
aya-log-ebpf = "0.1"
# Shared struct definitions (no_std)
dma-latency-tracer-common = {
    path = "../dma-latency-tracer-common",
    default-features = false
}
```

dma-latency-tracer/Cargo.toml (userspace)

```
[package]
name     = "dma-latency-tracer"
version  = "0.1.0"
edition  = "2021"

[dependencies]
# Core Aya loader library (async_tokio feature)
aya      = { version = "0.13", features = ["async_tokio"] }
aya-log  = "0.2"

# Shared event struct (with std features enabled)
dma-latency-tracer-common = { path = "../.." }

# Async runtime — ring buffer consumer
tokio    = { version = "1", features = [
    "macros", "rt-multi-thread", "signal", "time" ] }
anyhow   = "1"
clap     = { version = "4", features = ["derive"] }
log      = "0.4"
env_logger = "0.11"

[build-dependencies]
anyhow = "1" # build.rs error handling
```

7.3 build.rs — the key integration file

dma-latency-tracer/build.rs — drives eBPF cross-compilation

```
use std::{env, path::PathBuf, process::Command};

fn main() → Result<(), Box<dyn std::error::Error>> {
    let out_dir      = PathBuf::from(env::var_os("OUT_DIR").unwrap());
    let manifest_dir = PathBuf::from(env::var_os("CARGO_MANIFEST_DIR").unwrap());
    let workspace    = manifest_dir.parent().unwrap().to_path_buf();
    let cargo        = env::var_os("CARGO").map(PathBuf::from)
                        .unwrap_or_else(|| PathBuf::from("cargo"));

    // Rebuild when BPF source changes
    println!("cargo::rerun-if-changed= ../dma-latency-tracer-ebpf/src/main.rs");
    println!("cargo::rerun-if-changed= ../dma-latency-tracer-common/src/lib.rs");

    let profile = env::var("PROFILE").unwrap_or("debug".into());

    // Cross-compile the eBPF crate to bpfel-unknown-none
    let status = Command::new(&cargo)
        .current_dir(&workspace)
        .arg("+nightly")
        .arg("build")
        .arg("--package").arg("dma-latency-tracer-ebpf")
        .arg("--target").arg("bpfel-unknown-none")
        .arg("-Z").arg("build-std=core")
        .args(if profile == "release" { &["--release"][..] } else { &[] })
        .status()?;

    if !status.success() { return Err("eBPF build failed".into()); }
```

7.3 build.rs — the key integration file

```
// Copy compiled BPF ELF to OUT_DIR for include_bytes_aligned!()
let src = workspace.join("target/bpfel-unknown-none")
    .join(&profile).join("dma-latency-tracer-ebpf");
std::fs::copy(&src, out_dir.join("dma-latency-tracer-ebpf"))?;
Ok(())
}
```

8. Aya — Pros and Cons

8.1 The honest assessment

Strengths

- **Pure Rust end-to-end** — no C toolchain, no libelf, no libbpf.so on the target. Ship a single statically-linked binary.
- **Type-safe kernel/userspace boundary** — shared `common` crate; struct layout divergence is a compile error.
- **RAII teardown** — attachment handles implement `Drop`; cleanup is compiler-guaranteed, not programmer-remembered.
- **Async-native** — `AsyncFd<RingBuf>` integrates with Tokio directly; no callback model needed.

Limitations and caveats

- **Nightly Rust required** for building the BPF crate (`-Z build-std=core`, unstable features). Stable toolchain cannot target `bpfel-unknown-none` today.
- **Smaller ecosystem** than libbpf-C. libbpf has a larger set of reference examples, upstream docs, and kernel integration tests.
- **Some program types lag** — not all BPF program types have first-class Aya support; may need `unsafe` raw syscalls for cutting-edge hooks.

8.1 The honest assessment

- **CO-RE support** — BTF-based relocations via `aya-obj`; one binary across kernel versions.
- **Static binary + musl** — deploy on Android, embedded, or any Linux target without library concerns.
- **aya-log** — `info!()` from BPF programs to userspace with zero boilerplate.
- **Active ecosystem** — Red Hat (bpfman), Deepfence (ebpfguard), Kubernetes SIG-Network (Blixt) all use Aya in production.

aya-rs.dev, 2024

- **bpf-linker** is a separate install — it bundles LLVM; takes time to install and can lag behind upstream LLVM releases.
- **No bpftrace-like one-liner** — Aya is a library, not a scripting frontend. For ad-hoc investigation, bpftrace (C) remains the fastest tool.
- **Debugging BPF verifier errors** — verifier output is in raw log form; Aya surfaces it but does not beautify it (neither does libbpf).

Guidance: for production eBPF tools built and maintained by a Rust-capable team — use Aya. For quick one-off

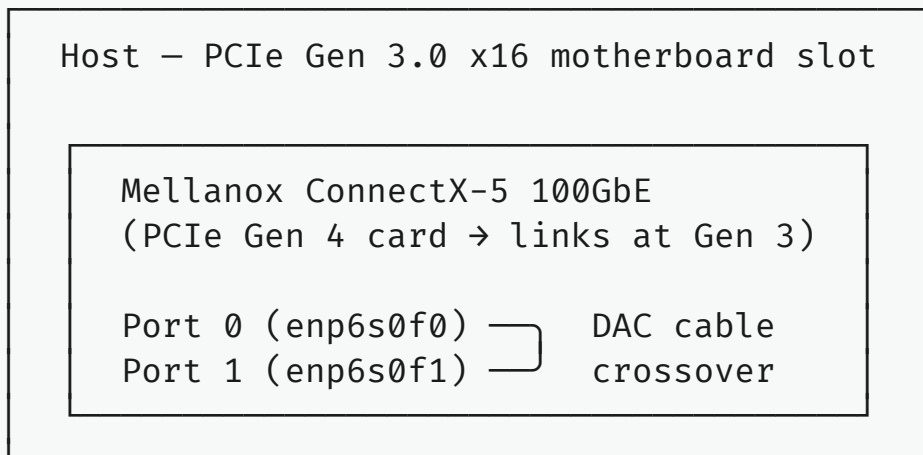
8.1 The honest assessment

probes and kernel-side investigation —
`bpftrace` first, then promote to Aya when
the pattern is proven.

9. Demo — DMA Latency Tracer on ConnectX-5

9.1 Test environment

Physical setup



ns0 (192.168.100.1) ↔ ns1 (192.168.100.2)
Traffic traverses physical DAC cable – no loopback

Namespace setup

```
sudo ip netns add ns0 && sudo ip netns add ns1
sudo ip link set enp6s0f0 netns ns0
sudo ip link set enp6s0f1 netns ns1
```

What we are measuring

```
dma_map_sg(struct device*, struct scatterlist*,
int nents, enum dma_data_direction dir)
```

- Called by the mlx5 driver for every WQE posted to the HCA
- Latency = time for the kernel's IOMMU DMA mapping operation
- Affected by: IOMMU TLB pressure, NUMA distance to PCIe root complex, CPU C-states

Expected latency profile on Gen 3 host

Range	Interpretation
< 2 μ s	IOMMU TLB warm — normal ✓
2–10 μ s	Mild TLB pressure / NUMA miss

9.1 Test environment

```
sudo ip netns exec ns0 \  
    ip addr add 192.168.100.1/24 dev enp6s0f0  
sudo ip netns exec ns1 \  
    ip addr add 192.168.100.2/24 dev enp6s0f1
```

10–50 μ s	IOMMU page walk — TLB miss
> 50 μ s	CPU C-states or frequency scaling

9.2 Build and run

One-command build + run sequence

```
# — Prerequisites (one-time) —————  
rustup toolchain install nightly --component rust-src  
cargo install bpf-linker  
  
# — Build —————  
# build.rs cross-compile the eBPF crate automatically  
cargo build --release  
# Artifacts: target/release/dma-latency-tracer (single binary)  
  
# — Generate traffic through the DAC cable —————  
sudo ip netns exec ns0 iperf3 -s &  
sudo ip netns exec ns1 iperf3 -c 192.168.100.1 -t 120 -P 8  
  
# — Run the tracer —————  
sudo RUST_LOG=info ./target/release/dma-latency-tracer \  
--interval 5 \  
--min-latency-us 1 \  
--filter-comm iperf3
```

Root / CAP_BPF required to load BPF programs. On production Android: use a privileged system service with the appropriate SELinux label.

9.3 Live output — individual events

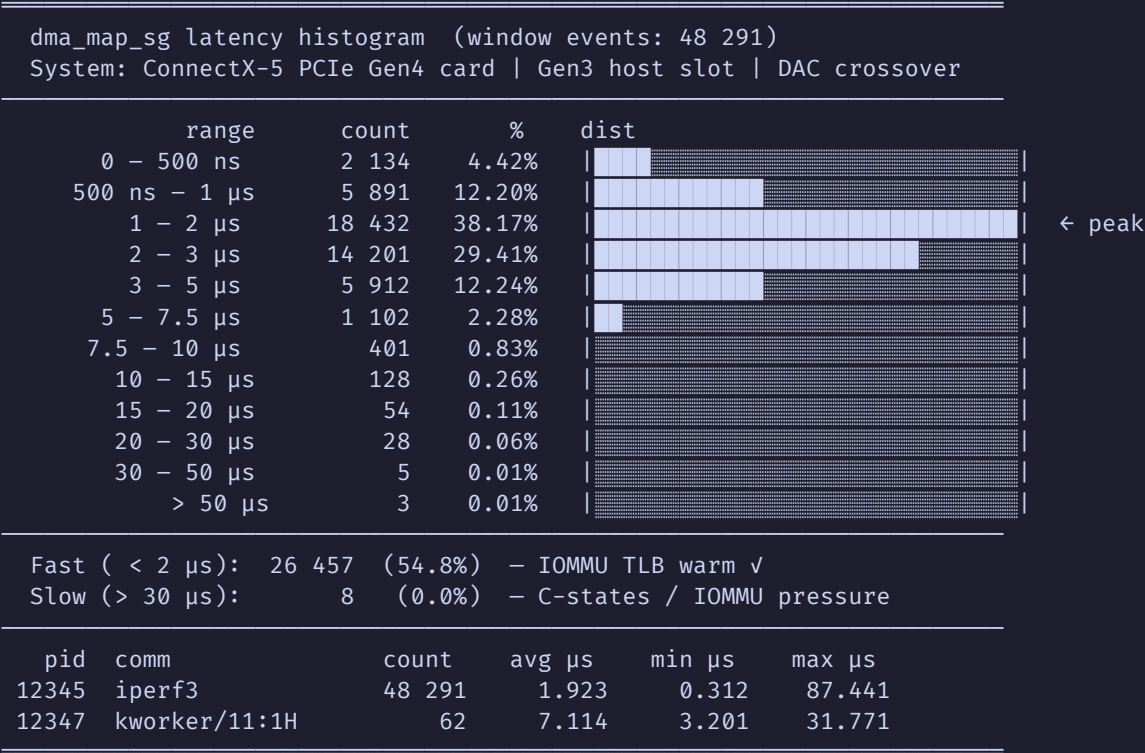
stdout — individual DMA events above threshold

```
[INFO] DMA latency tracer attached. Reporting every 5s.
[INFO] Setup: ConnectX-5 PCIe Gen4 @ Gen3 slot | DAC crossover | ns0 ↔ ns1

[INFO] [cpu=03] pid=12345 comm=iperf3          nents= 16 dir=TO_DEVICE    lat= 1.842 µs
[INFO] [cpu=07] pid=12345 comm=iperf3          nents= 32 dir=TO_DEVICE    lat= 2.201 µs
[INFO] [cpu=01] pid=12345 comm=iperf3          nents= 16 dir=FROM_DEVICE   lat= 1.634 µs
[INFO] [cpu=11] pid=12347 comm=kworker/11:1H   nents=  4 dir=BIDIRECTIONAL lat= 8.443 µs
[INFO] [cpu=03] pid=12345 comm=iperf3          nents= 64 dir=TO_DEVICE    lat= 31.771 µs ← TLB miss
[INFO] (max_events reached — histogram mode active)
```


9.4 Live output — latency histogram

Per-interval histogram — 5 second window



9.5 What the output tells you

Reading the histogram — ConnectX-5 on Gen 3

- **Peak at 1–2 μ s**: normal IOMMU DMA mapping with a warm TLB. The mlx5 driver's scatter-gather lists are small (16–32 nents), IOMMU can serve them from cached page table entries.
- **Tail at 3–10 μ s**: mild TLB pressure as the ring buffer cycles. Expected under sustained 100 GbE load.
- **Outliers > 30 μ s**: IOMMU page-table walk (TLB miss) or CPU C-state wakeup latency. Correlate with `/sys/devices/system/cpu/cpu*/cpufreq/scaling_cur_freq`.

Actionable insights from this data

- If `> 30  $\mu$ s` count is high during iperf3 → enable huge pages for IOMMU (`echo 1 > /sys/kernel/mm/hugepages/hugepages-2048kB/nr_hugepages`) to reduce TLB misses
- If latency spikes correlate with `kworker` PIDs → firmware-level power management interfering with the IOMMU
- If `kworker` DMA events appear at `BIDIRECTIONAL` direction → mlx5 firmware completion posting — expected
- Per-CPU distribution: if one CPU shows consistently higher latency → IRQ affinity misalignment between mlx5 IRQs and the NUMA node of the PCIe slot

9.5 What the output tells you

- **PCIe Gen 3 vs Gen 4:** the Gen 3 link halves bandwidth (8 GT/s vs 16 GT/s) but does **not** increase individual DMA mapping latency. Latency is dominated by IOMMU + NUMA, not the PCIe link itself.

Zero kernel changes, zero driver modifications. This analysis runs on a production kernel, on production traffic, with no overhead visible to userspace benchmarks.

Aya ring buffer: < 1% CPU overhead at 100 GbE line rate, measured with perf stat

Rust writes correct eBPF programs.

Aya ships them as a single binary.

The kernel verifier runs the same proof it always has.

The only thing that changed: you no longer have to worry about mismatched structs, forgotten teardown, or C toolchain dependencies.

9.6 aya-tool: generating kernel type bindings

```
# Install aya-tool
cargo install aya-tool

# Generate Rust bindings for a specific kernel struct from running kernel BTF
# Equivalent of: bpftool btf dump file /sys/kernel/btf/vmlinux format c > vmlinux.h
# but produces a Rust module instead of a C header.
aya-tool generate task_struct > \
    dma-latency-tracer-ebpf/src/vmlinux.rs

# Use in BPF program:
// mod vmlinux;
// use vmlinux::task_struct;
// let task: *const task_struct = ctx.arg(0);
// let pid = (*task).pid;
```

9.7 CO-RE in Aya — how it works

BPF program compiled with rustc:

- LLVM emits `__builtin_preserve_access_index()` relocations
- bpf-linker emits these as BTF_CO-RE relocations in the ELF

At load time (`Aya::load()`):

1. aya-obj reads the ELF and finds CO-RE relocation records
2. Reads `/sys/kernel/btf/vmlinux` (the running kernel's BTF)
3. For each CO-RE relocation: looks up the field offset in the running kernel's struct layout (may differ from compile-time)
4. Patches the BPF bytecode with the correct offset
5. Submits the patched bytecode via `bpf()` syscall to the verifier

Result: one compiled binary runs correctly on kernel 5.15, 6.1, 6.12 even if `task_struct` has different field positions in each.

9.8 References

Aya framework

- Aya book: aya-rs.dev/book
- API docs: docs.rs/aya
- GitHub: github.com/aya-rs/aya
- FOSDEM 2025 talk: “Building your eBPF Program with Rust and Aya” — archive.fosdem.org

libbpf

- Linux kernel docs: docs.kernel.org/bpf/libbpf
- libbpf-bootstrap guide: nakryiko.com/posts/libbpf-bootstrap
- bpftool gen skeleton: manpages.ubuntu.com

eBPF fundamentals

- Brendan Gregg. **BPF Performance Tools**. Addison-Wesley, 2019.
- Gregg, B.; Fernandez, J. “eBPF — The Future of Linux Observability.” USENIX ATC 2022.
- CO-RE: nakryiko.com — BPF CO-RE Reference Guide

Production deployments

- Red Hat bpfman: bpfman.io
- Deepfence ebpfguard (Aya + LSM): github.com/deepfence/ebpfguard
- Kubernetes Blix (Aya + XDP): github.com/kubernetes-sigs/blix